SIH 2026 · PS 26147 · NTRO

Project SANJAYA

The winning plan: a blind-signal forensics workbench that **proves** every answer it gives

> **Why the name.** In the Mahabharata, Sanjaya sees a distant battlefield he is not present at and reports it truthfully to the king. That is exactly the job: look at a recording from a sensor far away, with no cooperation from the transmitter, and report what it really is - with evidence. (Working name; team can change it.)

Built from: *introduction to 147.pdf* (what the PS asks) + *repos.pdf* (what 43 competing public repos already do) + a literature sweep of 2025-26 research on blind code/interleaver/scrambler identification, RF foundation models, open-set & conformal recognition, and wideband multi-signal detection.

Rule of this document: every feature is either (a) demanded by the PS, or (b) something the competing field does not do. Anything generic that 20+ teams already have is treated as table stakes, not as a USP.

1. The winning thesis

After reading 43 competing repos, the pattern is clear. Nearly every team builds the same **forward pipeline**: load file -> classify modulation -> demodulate -> try a fixed catalogue of decoders (K=7 Viterbi, RS(255,223), one LDPC) -> search for CCSDS sync word -> show a confidence %. The best ones add honesty labels ('UNKNOWN', 'VALIDATED').

Everyone else answers: "what do we think this signal is?"
SANJAYA answers: "here is the signal rebuilt from our decoded bits - subtract it from your recording and only noise is left."

We turn signal analysis from *classification* into *analysis-by-synthesis with proof*. Every hypothesis (modulation, symbol rate, interleaver, code, scrambler, frame) is only accepted when re-synthesising the waveform from the recovered bits explains the recorded energy down to the noise floor. Nothing is guessed; ambiguity is carried forward as a hypothesis and killed by evidence.

Three pillars, in priority order (exactly your criteria):

Pillar	What it means	Why it wins
A. Expected solution, 100%	Every clause of the PS (i-v + spectral features + GUI + header/payload) implemented, benchmarked, and traceable to a module.	Judges first tick the checklist. Missing one item loses to average teams. We make coverage visible with a live compliance matrix.
B. USPs nobody has	8 capabilities that 0-2 of the 43 repos attempt, each backed by 2025-26 research (Section 4).	Differentiation at the level of a professional COMINT tool, not a student dashboard.
C. Provable accuracy	SANJAYA-Bench: 10k+ ground-truth files, real over-the-air recordings, BER curves overlaid on theory, published KPIs.	'Most accurate' becomes a measured claim, not an adjective. NTRO engineers trust curves, not slogans.

One-line pitch

"Drop in any .IQ or .wav from any sensor. SANJAYA blindly recovers the complete transmitter recipe - modulation, symbol rate, interleaver, code, scrambler, frame format - names the standard if it knows it, reveals hidden signals underneath, and issues a Decode Certificate proving every parameter."

2. Competitive gap map (from repos.pdf)

Approximate counts from the 43 README/file-tree reviews. Green rows are where SANJAYA plants its flag.

Capability	Repos doing it (of 43)	SANJAYA
File loader + PSD + waterfall + constellation	~40	Table stakes (done better, not sold as USP)
ML modulation classifier (RF / CNN)	~25	Table stakes; we fuse 3 experts + conformal sets
Fixed catalogue decoders (K=7 Viterbi, RS 255/223)	~20	Table stakes; we identify codes blindly
GF(2) rank blind interleaver detection (hard bits)	~5	Upgraded to soft-decision + joint code/interleaver
Soft-decision (LLR) code & interleaver identification	0-1	USP
Open-set LDPC reconstruction (recover H, not pick from list)	0	USP
Blind scrambler (LFSR) polynomial recovery	~2 (fixed descramblers)	USP
Waveform-level re-synthesis residual verification	0	Core USP
Named-standard recognition (STANAG, CCSDS, AIS, NAVTEX...)	0	USP
Sample-rate back-solving + sensor clock-ppm forensics	0	USP
Protocol field discovery: counters, CRC polynomial, schema export	~1 (CRC search)	USP
Multi-signal scene + successive cancellation of decoded emitters	~3 partial (detection only)	USP
HF channel realism (Watterson) + adaptive equaliser	~1-2	USP

Capability	Repos doing it (of 43)	SANJAYA
Validation on real over-the-air recordings	2	USP (broader, standards-rich)
Statistical coverage guarantee on answers (conformal)	0 (1 does multiple-test correction)	USP

3. Pillar A - Expected solution: zero gaps

Each PS requirement, the module that delivers it, how we prove it, and the acceptance KPI. KPIs are **targets we commit to measure on SANJAYA-Bench**, not claims yet.

PS requirement	SANJAYA module & method	Proof / KPI target
.IQ and .wav input (different formats, different sensors)	**Recording Forensics**: WAV (mono real / stereo IQ / 8-32 bit / float), raw IQ dtype + endianness sniffer (u8, i8, i16, i32, f16, f32, f64; LE/BE), SigMF/sidecar/filename metadata, DC & IQ-imbalance correction; ambiguous choices (IQ swap / spectral inversion) kept as hypotheses.	Correct dtype on 100% of bench formats; ambiguity always surfaced.
(i) Sampling frequency	File Fs from header/metadata; if absent -> normalised analysis, then **back-solve true Fs** once a standard/baud is recognised; flag WAV files whose header Fs contradicts the anti-alias edge.	Fs recovered within 0.1% on standard-anchored files.
(i) Other parameters (feasible extras)	Centre/offset (M-th power FFT), occupied BW (99% & x-dB), SNR (M2M4 + spectral), symbol rate (cyclostationary lines + cyclic autocorr, refined in timing loop), roll-off, FSK tones/spacing/modulation index h, burst on/off times, sensor clock ppm.	Symbol-rate error < 0.05%; SNR +/- 1 dB above 0 dB.
(i) Modulation	3-expert fusion: physics features (cumulants C40/C42/C63, power-law lines, envelope & inst-freq stats) + compact IQ network + decode-fit expert (EVM after trial demod). Output = conformal **prediction set** + open-set 'novel' flag.	>= 98% top-1 at SNR >= 10 dB (BPSK..64QAM, 2/4/8-FSK, GMSK); 95% set coverage guaranteed.
(i) FEC	Blind code ID: conv (n,k,K, generators) via soft rank + dual-code/GJETP; RS (n,k,m) via Galois-field Fourier spectra; LDPC via catalogue syndrome test, then open-set sparse-dual recovery; concatenation found by recursive peeling.	Conv/RS ID >= 95% at channel BER 1e-2.
(i) Interleaving	Block (RxC), convolutional (B,J), diagonal, pseudo-random: soft rank-profile scan, joint with code ID; PN via standard permutations + LFSR seed search + parity-graph recovery.	Block/conv period ID >= 95% at BER 1e-2.
(ii) Demodulate FSK, QAM, PSK	Universal linear receiver: RRC matched filter, feed-forward (Oerder-Meyr) + Gardner timing, M-th power + Costas/DD-PLL carrier, CMA->DD-LMS fractionally-spaced equaliser; FSK filter-bank / discriminator; differential modes; **soft LLR output**.	BER within 0.5 dB of theory curve for each scheme (AWGN).
(iii) De-interleaving: Block, Conv, Diagonal, PN	All four implemented as invertible operators, parameterised by the identifier above; manual override in GUI.	Bit-exact round-trip tests; blind recovery on bench.
(iv) FEC: short-K conv + Viterbi, RS, concatenated, LDPC	C++ soft Viterbi (K 3-9, rates 1/2, 1/3, punctured 2/3, 3/4); RS Berlekamp-Massey (GF(2^m)); CCSDS-style concatenation; LDPC normalised min-sum (catalogue: CCSDS, 802.11n, DVB-S2 short, 5G NR BG2 + recovered H).	Coding gain matches theory within 0.5 dB; > 20 Mbit/s Viterbi.
(v) Bit-stream correlation, header & payload	**Bit-Lens**: autocorrelation frame length, soft sync-word discovery, aligned-frame raster, per-bit entropy map, field segmentation (constant / counter / CRC / length / payload), encryption-likelihood flag.	Frame length & sync word found on 100% of framed bench files.
Spectral features: Fs, constellation, waterfall	GUI: linked waterfall + PSD + constellation (time-animated) + eye + inst. frequency + cyclic spectrum + rank-profile plot.	All views on 100 MB file < 2 s.
GUI-based, automated	Desktop PySide6 + pyqtgraph; **Autopilot** (one click, full chain) and **Expert** mode (override any stage, re-run downstream).	Full chain on 1 M-sample file < 10 s.
'GNU Radio, python, C++'	Python orchestration; C++ (pybind11) hot kernels; GNU Radio OOT block exposing the analyser in flowgraphs.	All three named technologies used for a real reason.

4. Pillar B - The eight USPs

Format for each: what it is, why nobody in the field has it, the research behind it, how we build it with realistic effort, and the moment it creates in the demo.

USP 1 - Proof-of-Decode: re-synthesis residual verification (the core idea)	
What	After decoding, SANJAYA re-encodes the recovered bits through the discovered chain (code -> interleaver -> scrambler -> mapper -> pulse shape), applies the estimated channel (gain, CFO, phase, timing, equaliser taps), and **subtracts the rebuilt waveform from the recording**. It reports *explained energy %*, residual-to-noise ratio and a whiteness test. If the residual is white noise at the estimated noise floor, the whole recipe is proven. Output: a signed **Decode Certificate** (file hash, every parameter, its evidence, its confidence set).
Why nobody has it	Best competitors verify at bit level (CRC / re-encode BER). Nobody verifies at the waveform level, which is the only check that also validates symbol rate, pulse shape, channel and modulation at once - and catches 'plausible but wrong' decodes.
Research basis	Analysis-by-synthesis and successive interference cancellation are standard receiver theory; we apply them as a forensic verifier.
Build / effort	Medium. Reuses our own modulator (needed for the generator anyway). 1 person, ~1.5 weeks.
Demo moment	*Show the waterfall, press 'Prove': the signal vanishes from the waterfall, leaving only noise. Certificate prints.*

USP 2 - Soft-decision, joint blind coding-layer engine	
What	Identifies interleaver + code **jointly** from soft LLRs instead of hard bits: rank-profile scans on soft matrices, dual-code / GJETP recovery of convolutional generators, Galois-field Fourier spectra for RS, recursive peeling for concatenated codes, and an **open-set LDPC mode** that reconstructs a sparse parity-check matrix for short codes instead of choosing from a list. The GUI shows the **rank-profile plot** - the visible dip at the true interleaver period - so the analyst sees why.
Why nobody has it	~5 repos do hard-decision GF(2) rank on one layer; everyone else uses a fixed decoder list. Soft, joint and open-set identification: none.
Research basis	Subspace-coding blind code ID (arXiv 2601.15903, 2026); open-set non-binary LDPC recognition via GJCE + sparsification (Digital Comms & Networks, 2025); CNN / ConvLSTM-TFN soft blind conv-code recognition (Int. J. Intelligent Systems 2025; Sensors 2025); dual-branch CNN channel-code recognition (Sci. Reports 2026); joint interleaved-conv-code recovery via parity-check graphs (arXiv 1501.03715); GFFT + reliability verification for RS (IEEE 2023).
Build / effort	High, but staged: hard rank (week 3) -> soft rank (week 5) -> open-set LDPC for n <= 1024 (week 7-8). 2 people. Bit-packed uint64 GF(2) elimination in C++.
Demo moment	*Judge picks interleaver size secretly; rank-profile dips exactly there in seconds.*

USP 3 - Blind scrambler recovery	
What	Additive (synchronous) and self-synchronising LFSR scramblers identified blindly (Walsh-Hadamard / eigen-analysis, Berlekamp-Massey on idle or preamble runs), plus a library of standard polynomials (CCSDS, DVB PRBS-15, 802.11 x^7+x^4+1, V.35, ...). Descrambling is inserted into the chain automatically.
Why nobody has it	Real systems scramble between FEC and modulation; the PS does not mention it, so almost every team will fail on real signals. ~2 repos have a fixed descrambler; none identify an unknown one.
Research basis	Joint blind identification of cyclic codes and synchronous scramblers with soft information (IET Communications, 2026); feedback-polynomial reconstruction in noise (IET Comms 2022-23).
Build / effort	Medium. 1 person, ~1 week.
Demo moment	*'Other tools stop here - bits still look random. SANJAYA finds x^15+x^14+1 and the frames appear.'*

USP 4 - Signal Genome: named-standard recognition + sample-rate back-solving	
What	A descriptor library (~30 real standards to start: STANAG 4285/4529/4539, MIL-STD-188-110A/B serial, CCSDS TM, DVB-S, AIS, NAVTEX/SITOR-B, RTTY, POCSAG, ACARS, HFDL, DMR, dPMR, TETRA, NOAA-APT, APRS/AX.25, ...). Measured recipe vectors are matched to it: *"91% consistent with STANAG 4285, 600 bps long interleave"*. Once a standard is matched, its nominal baud gives the **true sample rate** of a headerless IQ file and the **sensor clock error in ppm**.
Why nobody has it	0 of 43. Only commercial suites (PROCITEC, WAVECOM) do this - it is what makes analysts' output directly usable by 'designated sensors'.
Research basis	Public standards documents + Signal Identification Wiki descriptors; open-set recognition literature (Open-DS, IET RSN 2025) for 'close to X but not X'.
Build / effort	Low-medium: data entry + scoring. 1 person part-time throughout. Generators for ~10 standards double as test data.
Demo moment	*Real HF recording loads -> 'STANAG 4285 detected, recorder clock +38 ppm fast'.*

USP 5 - Bit-Lens: protocol structure discovery and schema export	
What	Beyond sync words: aligned-frame raster with adjustable width (the classic analyst view), per-bit entropy / constancy heat-map, automatic detection of **counters** (increment patterns), **length fields**, and **CRC fields including the polynomial** (bits that are a linear GF(2) function of others -> solve). High-entropy payload flagged as likely encrypted/compressed. Exports the discovered frame layout as a **Kaitai Struct / Wireshark-dissector** style schema.
Why nobody has it	Everyone stops at 'sync word found at offset N'. The PS explicitly asks to identify header and payload - we hand back a dissector.
Research basis	Deep-learning blind frame-sync recognition (Sensors 2024); bitstream protocol format reverse-analysis and hierarchical format inference (Empirical SE 2026).
Build / effort	Medium. 1 person, ~1.5 weeks. Pure GF(2) linear algebra + statistics; no ML needed.
Demo moment	*Bit raster snaps into columns; fields colour themselves: SYNC | COUNTER | LEN | PAYLOAD | CRC-16/CCITT.*

USP 6 - Scene mode: multi-signal detection + successive cancellation	
What	Captures from real sensors hold many emitters. OS-CFAR on the spectrogram + connected components list every emitter (freq, BW, on/off time); each is channelised and analysed. After a strong signal is decoded and proven (USP 1), it is **subtracted** - revealing weaker signals hidden under it, which are then analysed in turn.
Why nobody has it	~3 repos detect multiple signals; none decode-and-cancel to uncover buried ones.
Research basis	Wideband multi-signal detection (CSRD2025 dataset, arXiv 2508.19552; ZoomSpec physics-guided coarse-to-fine sensing, arXiv 2604.13568; dense multi-emitter blind detection, 2026); TorchSig WidebandSig53 benchmark.
Build / effort	Medium. CFAR version first (week 4); optional YOLO-style detector trained on TorchSig later.
Demo moment	*Strong QPSK is removed, a 12 dB weaker FSK beacon 'appears' in the residual waterfall and gets decoded.*

USP 7 - HF-grade robustness (the PS names HF first)	
What	Watterson/ITU-R F.1487 HF channel simulator (multipath delay + Doppler spread; good / moderate / poor profiles) in the generator and bench; fractionally-spaced adaptive equaliser; SSB-audio .wav handling (real pass-band at ~1-3 kHz, spectral inversion USB/LSB kept as a hypothesis). Validation on **real public HF recordings** (KiwiSDR/WebSDR: NAVTEX, RTTY weather, STANAG 4285, HFDL, ALE).
Why nobody has it	Almost every repo tests on AWGN only. HF fading breaks naive timing/carrier loops - they will fail the moment a judge brings a real HF file.
Research basis	Standard HF channel models; DenoMAE multimodal denoising for low-SNR modulation signals (arXiv 2501.11538) as optional front-end.
Build / effort	Medium. Simulator ~3 days; equaliser ~1 week; data collection ongoing.
Demo moment	*Same STANAG signal through 'poor' HF channel: competitor-style receiver fails live, SANJAYA locks.*

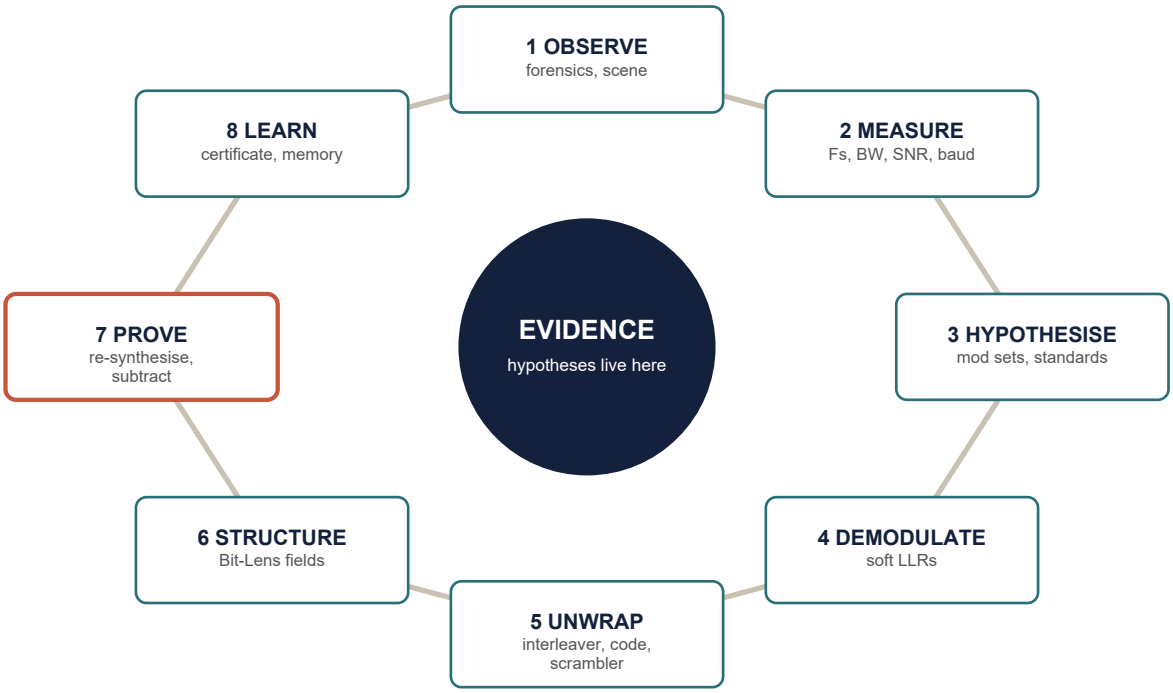
USP 8 - Trust layer: conformal sets, open-set 'novel', few-shot teach	
What	Classifiers output **conformal prediction sets** with a guaranteed coverage (e.g. {QPSK, 8PSK} at 95%), not a naked %. An embedding-prototype head flags **novel** signals and lets an analyst **teach** a new class from ~5 examples; the lab's confirmed results become a local 'mission memory' that recognises previously seen emitters.
Why nobody has it	Repos print arbitrary confidence percentages or ad-hoc 'epistemic states'. None give a statistical guarantee or learn from the analyst.
Research basis	Conformal prediction for wireless AI / AMC (Conformal Shield); RF foundation models & few-shot adaptation: IQFM (arXiv 2506.06718, 2025), EMind (arXiv 2508.18785, 2025), Radio-FM (arXiv 2608.05793, 2026), RFPrompt (arXiv 2605.03279, 2026), geometry-aware contrastive few-shot AMR (arXiv 2605.26600, 2026).
Build / effort	Low for conformal (split-CP is ~50 lines on a calibration set); medium for few-shot (small contrastive encoder trained on TorchSig + our generator).
Demo moment	*Unknown chirp signal -> 'NOVEL, closest: none'. Analyst labels 5 -> next file recognised.*

Deliberately NOT doing (anti-generic list)

- **LLM chatbot / 'AI copilot'** - at least 4 repos have one; for an air-gapped intelligence user it is a liability, not a feature. Our explanations come from evidence, not generated text.
- **Cloud, login, RBAC, Firebase, Postgres** - irrelevant to NTRO's offline analyst; costs weeks.
- **Neon 'military-grade' dashboards and marketing adjectives** - judges from NTRO will discount them. Our UI is a quiet, dense analyst instrument.
- **Black-box-only deep learning** - DL is one of three experts, never the sole judge.

5. The SANJAYA analysis cycle

The product itself is a loop, not a line. Each turn either proves a hypothesis or kills it; every stage can hand ambiguity forward.



Failed proof at 7 -> back to 3 with the next-best hypothesis. Proven signal at 7 -> subtracted -> scene re-scanned for hidden emitters.

Stage	Key algorithms	Hand-off (what it passes on)
1 Observe	dtype/endianness sniff, SigMF/metadata, DC/IQ-imbalance, OS-CFAR scene detection, channelisation	List of emitters + normalised complex baseband per emitter + unresolved format hypotheses
2 Measure	Welch PSD, 99% BW, M2M4 SNR, cyclostationary baud, M-th power CFO, inst-freq histogram, burst timing	Parameter vector with uncertainty intervals
3 Hypothesise	Cumulant rules + IQ network + standards match -> conformal set	Ranked, pruned hypothesis list (mod x baud x standard)
4 Demodulate	RRC MF, Oerder-Meyr/Gardner, Costas/DD-PLL, CMA->LMS FSE, FSK bank	Soft LLR stream per hypothesis incl. phase rotations
5 Unwrap	Soft rank profile, joint interleaver+code ID, Viterbi/RS/LDPC, recursive peel, LFSR recovery	Decoded bits + recipe + syndrome/CRC evidence
6 Structure	Autocorr frame length, sync discovery, counters, CRC polynomial solve, entropy	Frame schema (Kaitai-style) + header/payload split
7 Prove	Re-encode -> remodulate -> apply channel -> subtract -> whiteness test	PASS/FAIL + explained-energy % + residual for next scene pass
8 Learn	Certificate PDF/JSON, mission memory, few-shot prototype update	Report for 'designated sensors'; faster recognition next time

Software layout (kept deliberately small)

Layer	Tech	Why
Core DSP & coding	Python 3.11 + NumPy/SciPy; **C++17 via pybind11** for Viterbi, LDPC min-sum, bit-packed GF(2) elimination; *galois* for GF(2^m) RS math	Speed where it matters; Python everywhere else for velocity
Reference checks	AFF3CT / liquid-dsp / GNU Radio blocks used as **cross-check oracles** in tests	Proves our decoders match established implementations
ML	PyTorch small encoders (< 5 M params, CPU inference), scikit-learn for feature expert, split conformal	Runs offline on a laptop
GUI	PySide6 + pyqtgraph (OpenGL), one window, docked panels	Handles 100 MB+ files interactively; true desktop tool

Layer	Tech	Why
Integration	CLI for batch; **GNU Radio OOT Python block**; SigMF annotations out	Fits NTRO's existing sensor chains - the PS names GNU Radio
Packaging	Single offline installer (PyInstaller) + wheels; no network access at runtime	Air-gap ready

6. Implementation sequence (the build cycle)

Ten weeks, six phases. **Rule: main branch is demo-able at the end of every phase** - if time runs out, we present the last green phase, never a broken build. Every sprint ends with the SANJAYA-Bench run and a KPI diff.

Phase / weeks	Build	Exit criterion (must be true to move on)
P0 - Idea round (days 1-4)	PPT with thesis, gap map, architecture, USPs; 2-minute prototype video: synthetic QPSK+conv+block file decoded and 'proved' (residual vanishes).	Idea submitted with a **working** proof-of-decode clip - most idea decks will have only mockups.
P1 - Spine (W1-2)	Recording Forensics; SANJAYA-Gen v1 (7 mods, RRC, CFO, timing, AWGN, truth JSON); measurement suite; linear receiver for BPSK/QPSK + FSK; GUI shell (waterfall, PSD, constellation); bench harness + CI.	Synthetic BPSK/QPSK/2FSK -> bits end-to-end in GUI; bench prints KPIs automatically.
P2 - Coding core (W3-4)	C++ soft Viterbi; RS (galois); block + conv interleavers with hard rank-profile ID; scrambler library + LFSR recovery; sync discovery; Bit-Lens v1 raster; **Proof-of-Decode v1**; OS-CFAR scene mode.	Blind recovery of QPSK + scrambler + block interleaver + K=7 code, proven by residual.
P3 - Full PS coverage (W5-6)	8PSK, 16/64-QAM, GMSK, 4/8-FSK; equaliser; diagonal + PN interleavers; concatenated RS+conv (recursive peel); LDPC catalogue + min-sum; 3-expert AMC + conformal sets; soft rank-profile; Watterson HF channel in bench.	**Compliance matrix 100% green** on bench (every row in Section 3).
P4 - USP depth (W7-8)	Signal Genome (30 standards) + Fs/ppm back-solve; open-set LDPC (n <= 1024); Bit-Lens v2 (counters, CRC polynomial, schema export); SIC hidden-signal reveal; few-shot teach; GNU Radio block; real OTA dataset (>= 30 recordings).	Each USP has a 30-second scripted demo that works 10/10 times.
P5 - Hardening (W9)	10k-file bench; BER-vs-theory curves; failure triage; performance (C++ hot paths, mmap streaming); cut anything below 90% reliability.	KPI table filled with measured numbers; no crash on 500 random files incl. garbage input.
P6 - Showtime (W10)	Mystery-signal challenge app; Decode Certificate polish; pitch deck with real curves; judge Q&A drills; offline installer on 2 laptops + USB backup.	Full demo rehearsed 5 times under 7 minutes, including one live failure-recovery.

Team of six - ownership, not overlap

Role	Owns	Phase peak
1. DSP lead	Forensics, measurements, receiver, equaliser, HF channel	P1-P3
2. Coding-theory lead	Viterbi/RS/LDPC, rank engine, joint ID, open-set LDPC	P2-P4
3. Protocol & crypto-structure	Scramblers, Bit-Lens, CRC solve, schema export, Signal Genome library	P2-P4
4. ML & trust	3-expert AMC, conformal, open-set, few-shot, dataset pipelines (TorchSig/RadioML/own)	P3-P4
5. GUI & integration	PySide6 app, Autopilot/Expert modes, GNU Radio block, installer	P1, P4-P6
6. Bench, data & pitch (team lead)	SANJAYA-Gen, bench/CI, OTA recordings, KPIs, deck, demo script	all

7. Pillar C - The accuracy programme

'Most accurate system' is only believable with numbers nobody else shows. SANJAYA-Bench is built in week 1 and runs on every merge.

Component	Content
SANJAYA-Gen (synthetic)	Standards-accurate generators; mods x baud x SNR (-5..30 dB) x CFO x clock drift x HF Watterson profiles x code x interleaver x scrambler x frame; sensor impairments (DC, IQ imbalance, dtype, sample-rate mismatch). Every file ships with truth JSON. Target 10k+ files.
Public ML datasets	TorchSig Sig53 / WidebandSig53 (pre-training, multi-signal), RadioML 2018.01A (comparability with published AMC numbers), CSRD2025 (wideband sensing).
Real over-the-air set	>= 30 recordings of public, unencrypted signals: HF via KiwiSDR/WebSDR (NAVTEX, weather RTTY, STANAG 4285 idle, HFDL, ALE); VHF/UHF via RTL-SDR (AIS GMSK, POCSAG, APRS AFSK, NOAA-APT). Only structure is analysed; no private content is decoded or shown.
Curves we will show	BER vs Eb/N0 per modulation with theory overlay; coding gain vs theory for each code; AMC confusion matrix per SNR; interleaver/code ID accuracy vs channel BER; conformal coverage vs target; false-decode rate (should be ~0 thanks to Proof-of-Decode).
Honesty metric	**False-certificate rate**: how often a PASS certificate is wrong. Target 0 on the whole bench. This is the number that sells the system to NTRO.

8. How we win the room

Seven-minute demo script

Min	Beat	What judges see
0:00	The problem, in one frame	A raw HF recording: noise-like waterfall. 'An analyst spends hours on this. Watch.'
0:40	Autopilot on a real HF file	Scene -> STANAG 4285 recognised -> sample rate & clock ppm back-solved -> constellation locks through fading.
2:00	The Mystery-Signal Challenge	A judge picks modulation, interleaver size, code and scrambler on our generator tablet, blind to us. SANJAYA recovers all of them; rank-profile dip shown.
3:30	Proof-of-Decode	Press 'Prove': signal disappears from the waterfall, residual is white. Certificate prints.
4:30	Hidden signal	After cancellation, a weak FSK beacon emerges from under the strong carrier and is decoded.
5:30	Bit-Lens	Raster snaps into columns; fields auto-labelled SYNC | COUNTER | LEN | PAYLOAD | CRC-16 (polynomial shown); schema exported.
6:15	Numbers	One slide: bench size, BER-vs-theory, false-certificate rate 0, speed. Close on the one-line pitch.

Judge questions we must answer cold

Likely question	Our answer (short)
How do you know it's right on real data with no ground truth?	Proof-of-Decode: the rebuilt waveform explains the recording to the noise floor; plus CRC/syndrome; plus real OTA set where standards are public.
Isn't blind LDPC impossible?	For long codes, yes in general - we match catalogue codes by syndrome and reconstruct short codes (n <= 1024) with the 2025 GJCE + sparsification method; we say exactly where the limit is.
What if the signal is encrypted?	We still recover the full transmission recipe and frame structure; Bit-Lens flags the payload as high-entropy. Decryption is out of scope by design.
Why not just a deep-learning model?	DL is one expert of three. DL cannot give a coding-layer proof; physics + algebra can. Conformal sets make the DL part honest.
Will it run on our machines?	Offline installer, CPU-only, laptop-class; GNU Radio block drops into existing flowgraphs; SigMF in/out.
What happens with an unknown modulation?	Open-set flag 'NOVEL', nearest known shown, analyst can teach it from ~5 examples.

9. Risks and cut-lines

Risk	Mitigation	Cut-line if late
Soft joint code/interleaver ID takes longer	Hard-decision rank version lands first (P2) and already beats most of the field	Ship hard-decision + catalogue; show soft as benchmark plot

Risk	Mitigation	Cut-line if late
Open-set LDPC unstable	Restrict to short codes; catalogue path always available	Present as 'research mode' with honest limits
Real OTA recordings noisy / scarce	Start collecting in week 1; KiwiSDR is available 24x7	Keep 5 best-quality files for demo
HF equaliser convergence	Start with CMA->LMS; test on Watterson from P3	Limit to 'moderate' HF profile in claims
GUI performance on big files	pyqtgraph + decimated views + mmap from day 1	Cap interactive view, full analysis via CLI
Scope creep	Section 4 is the whole USP list - nothing new after W7	Drop USP 8 few-shot first, then USP 6 SIC

10. Research references (2025-26 focus)

Summarised from abstracts and publisher pages during the sweep; team members should read the full papers for the modules they own.

#	Paper	Link
1	Blind Identification of Channel Codes: A Subspace-Coding Approach (arXiv 2601.15903, 2026)	https://arxiv.org/abs/2601.15903
2	Open-set blind recognition of non-binary LDPC codes (Digital Communications and Networks, 2025)	https://www.sciencedirect.com/science/article/pii/S2352864825001506
3	Blind Recognition Algorithm of Convolutional Code via CNN (Int. J. Intelligent Systems, 2025)	https://onlinelibrary.wiley.com/doi/10.1155/int/3183819
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